

पत्रांक / No: 6346 निदे०(आप०)/Dir (Op)/SOP

दिनांक : 31.12.22

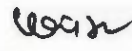
विषय :- परीक्षण एवं परिचालन कार्यों हेतु स्टेन्डर्ड ऑपरेटिंग प्रोसिजर लागू कराये जाने हेतु।

मुख्य अभियन्ता (पारेषण पश्चिम/दक्षिण-मध्य/दक्षिण-पश्चिम/दक्षिण-पूर्व/उत्तर-पूर्व/मध्य)
उ०प्र० पावर ट्रांसमिशन कारपोरेशन लिमिटेड,
मेरठ/झाँसी/आगरा/प्रयागराज/गोरखपुर/लखनऊ।

कृपया उपरोक्त विषयक इस कार्यालय के पत्र सं० 3694-निदे०(आप०)/SOP दिनांक 01.07.2022 का सन्दर्भ ग्रहण करना चाहें। जिसके द्वारा परीक्षण एवं परिचालन कार्यों हेतु स्टेन्डर्ड ऑपरेटिंग प्रोसिजर के निर्धारण हेतु अधीक्षण अभियन्ता, विद्युत परीक्षण एवं परिचालन मण्डल, लखनऊ, मेरठ, झाँसी एवं प्रयागराज की समिति द्वारा दी गयी संस्तुतियों आपके अवलोकनार्थ एवं सुझावों हेतु संलग्न कर प्रेषित की गयी थी।

कृपया समिति द्वारा की गयी संस्तुतियों के आधार पर परीक्षण कार्यों को कराना सुनिश्चित करें।

संलग्नक :- यथोपरि।


(पंकज मालवीय)
मुख्य अभियन्ता (सम्बद्ध)
निदेशक (आपरेशन)

प्रतिलिपि :-

निदेशक (आपरेशन), उ०प्र० पा० ट्रा० का० लि०, शक्ति भवन (वि०), लखनऊ को विनम्र सूचनार्थ।

SOP FOR TESTING AND COMMISSIONING RELATED WORKS

Testing and Commissioning is an important wing of Transmission which looks after specific tasks related to Routine Testing of Protection and Pre-commissioning Test of New Substation/ New Elements. Apart from this, T&C wing also maintains tele-protection signaling and performs pre commissioning checks on energy meters and assists in energy meter data collection through data collection devices (MRI). Detail of activities performed by Test and Commissioning wing is summarized in the table given below:

Sl. No.	Main Work/Activity with Time	Periodicity	Shut down required/Not required	Activities included in the work	Remarks
A 1.	<u>Routine Testing Transmission Lines</u> 765kV 2 days/bay 400kV 1-2 days/bay 220kV 1 day/bay 132kV 1-2 bays/day 33kV 5-7 bays/day	Yearly	Required	<u>Works to be carried out by T&C Wing-</u> Testing of all Protection relays distance and back up protection. Testing of master trip functionality, thorough Tripping and Auto Reclose, SOTF, Broken Conductor, PSB Functionality, Checking Healthiness of Carrier Inter-tripping System including CTx, CRx and DT. Checking of Time Synchronization Status, Event Logger functionality. <u>Works to be carried out by Transmission Wing-</u> Measurement of IR value of CTs, Testing of CB timings, checking of isolator contacts, oil level of CTs and leakage attend and checking of other Bay Equipment. Attending of Hot Spots identified in Thermo vision Scanning before Shut Down including replacement of damaged jumpers, Clamps etc. Checking of Earth Resistance of Delta Earthing of LAs of Lines. Checking of Healthiness of Main Earth Mat of Switch-Yard. Thorough Cleaning, Tightening of Cables/Wires/Contacts in CR Panels. Tightness of Bay Jumpers, Run Through Conductors, Connectors, Clamps etc. In case of SAS based substations complete checking of SAS system including status of breakers and isolators, Data Transmission, Healthiness of DTPC etc. also to get carried out by Transmission Wing.	Shut Down shall be arranged by Transmission Wing. As far as possible Shut down should be availed on returnable basis. During shutdown period Transmission Wing shall ensure to complete the necessary O&M works. Review of critical test results should be done at circle/ zone level

				Transformers. Attending of Hot Spots identified in Thermovision Scanning before Shut Down including replacement of damaged jumpers, Clamps etc. of Transformer/Reactors. Thorough Cleaning, Tightening of Cables/Wires/Contacts in CR Panels. Checking of Fire Fighting Equipment like Emulsifier System, NIFPS, Fire Detection System, Sand Barriers and Alarms. Checking of healthiness and working of Cooling Fans & Cooling Pump. Tightness of Bay Jumpers, Run Through Conductors, Connectors, Clamps etc. In case of SAS based substations complete checking of SAS system including status of breakers and isolators, Data Transmission etc. also to get carried out by Transmission Wing.	
3.	Capacitor Bank 132 KV Capacitor Bank 1 Day Per Capacitor Bank 33 KV Capacitor Bank 1 Day Per Two Capacitor Bank	Once in a Year	Required	<u>Works to be carried out by T&C Wing-</u> Testing of Under Voltage, Overvoltage, Overcurrent, Earth Fault and Unbalance Current Relays. <u>Works to be carried out by Transmission Wing-</u> Attending of Hot Spots identified in Thermovision Scanning before Shut Down including replacement of damaged jumpers, Clamps etc. and Thorough Cleaning, Tightening of Cables/Wires/Contacts in CR Panels. Tightness of Bay Jumpers, Run Through Conductors, Connectors, Clamps etc.	During shutdown period transmission wing shall ensure to complete the necessary O&M works.

[illegible]

C	<u>Carrier Communication</u>	Monthly	Not Required	<u>Works to be carried out by T&C Wing-</u> Measurement of Tx, Rx and guard signal level of all PLCC channels using Selective Level Meter and any necessary adjustment. Loop testing is done if any problem is detected during periodic level check. Checking of DC supply voltage and various regulated DC voltages in the PLCC Cabinet.	
D	<u>Commissioning of Substation/Augmentation of Bay</u>	As per requirement	Not applicable	All Equipment like Transformers, Bus and line Reactors, CTs, CVTs are tested. Control and Protection wiring is checked. All relay settings are programmed into respective relays and subsequently testing is carried out. Tripping matrix is verified. Testing of switch gears like DCRM and CRM is done. Commissioning of PLCC channels and protection couplers is carried out. Bus-bar and LBB schemes are tested for stability	In case the Commissioning/Augmentation is done on turnkey basis, all these functions will be performed by the contractor and T&C and Maintenance Engineer shall witness their respective activities.
E	<u>MRI Work</u>	Monthly	Not Required	<u>Works to be carried out by T&C Wing-</u> MRI of all T to D interface points, MRI of all open access consumers and generating stations as per Connectivity Agreement. JMR of large power plants. Uploading of all MRI data on SLDC portal.	Weekly MRI not possible as the same shall adversely affect the protection work.

				Correction of time drift in Energy Meters.	
F	<u>Energy Metering</u>	Yearly/As per requirement	Required some times	<u>Works to be carried out by T&C Wing-</u> Yearly checking of accuracy of open access point energy meters. Arrangement for Calibration of energy meter to be done by Transmission Wing. Witnessing the checking of accuracy of new open access connections carried out by NABL certified testing lab at site or at works Joint sealing of new open access connections as per connectivity agreement to be done by T&C Wing.	
G	<u>Testing and Replacement of defective outdoor/indoor equipment of different voltage level</u>	As per requirement	Required	Replacement of Equipment to be arranged by Transmission Wing. Testing of new transformer/CT/PT/CVT or other equipment before putting into service Checking of marshaling box of connections in case of replacement of transformer Checking of protection (if necessary) by T&C Wing.	
H	<u>DC System Battery</u>	Once in 3 Years		C-10 Test to be done by Transmission.	
I	<u>Tripping Analysis</u>	Monthly/SOS	Not Required	1. Analysis of tripping is performed on the basis of flags, DRs and event logs. 2. For all final trippings flash report, DR, event log and analysis are uploaded on online tripping monitoring system. 3. For major grid disturbances reporting is done in separate prescribed proforma and matter is represented in PSC (Protection Sub-committee) Meetings. 4. In case of erratic tripping DR file is played back (Transplay) on spare relay after configuring the spare relay identical to suspected relay.	

J	Energy Accounting	Monthly	N.A.	Energy account prepared by various transmission circles is compiled at T&C circle to prepare Zonal energy account.	
K	Checking of CVT/PT voltages of Substation	Monthly / SOS	Not required	PT/CVT voltages are measured to find out any drift in output voltage.	
L	Protection Audit of Sub-Station (Internal or Third Party) 765kV 4-5 days/substation 400kV 4-5 days/substation 220kV 3-4 days/substation 132kV 2-3 day/substation	Once in 4 years or as per instructions of higher authorities	Not Required	Checklist for Transmission lines and cables: 1) Independent Main-I and Main-II protection (of different make OR different type) is provided with carrier aided scheme and type of operating characteristics. 2) Are the Main-I & Main-II relays connected to two separate DC sources (Group-A and Group-B). 3) Zone-1, Zone-2, Zone-3 and Zone-4 settings. 4) Resistive reach for Ground and phase fault element to give maximum coverage. 5) Switch on to fault (SOTF) function and SOTF initiation implementation using hardwire logic. 6) In case of Breaker and half switching scheme, whether initiation of line SOTF from CB closing has been interlocked with the other CB. 7) VT fuse fail detection function to block the distance function operation on VT fuse failure. 8) In case of Cables, is unit protection provided as Main-I & Main-II protection with distance as back-up. 9) Two stages Over-Voltage protection provided for 765 & 400kV Lines- Stage-I: (typical value -106 to 112 %, delay: 4-7 Sec) Stage-II: (typical value -140 to 150%, delay: 0 to 100msec.) 10) 1-ph Auto -reclosing on 765, 400 & 220kV line set values:	Protection audit must be carried out by Officers other than those responsible for maintenance/ testing of that sub-station.

				a. Dead Time: (Typical 1 Sec) b. Reclaim Time: (typical 25 Sec) 11. Distance Communication Scheme Permissive Over Reach (POR) for short lines and Permissive Under Reach (PUR) for long lines. 12. Zones blocked on Power swing block function, Setting for Unblock timer (typical 02 second) Out of Step trip enabled or not. 13. Disturbance recorder and Fault locator, standalone or built in Main relay, Whether DR is having automatic fault record download facility to a central PC Whether DR is time synchronised with the GPS based time synchronizing equipment. 14. Whether DR digital channel as a minimum contain the CB status, Main-I & II trip status, LBB trip status, Over-voltage trip status, Stub protn trip status, Permissive and direct carrier receive status, Line reactor trip status. 15. Whether substation (765, 400, 220kV) is having Event logger facility (standalone or built-in-SAS). <u>Checklist for Power Transformers and Reactors-</u> 1) Separate DC sources for Group A and Group B protections of power transformers. 2) Duplicated PRD and Bucholtz initiating contact for power transformers at 765kV and 400kV levels. 3) In case of Breaker & half switching scheme, whether CT associated with Main & Tie Breakers are connected to separate bias winding of the low impedance Biased differential protection in order to avoid false operation due to dissimilar	
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				CT response. 4) Restricted earth fault (REF) protection used a high impedance type or low impedance type. 5) Are Main protection relays provided and directional over current & earth fault relays of Transformer are of numerical design. Do the back-up earth fault relays have harmonic restrain feature. 6) Fire Protection System- a. (HVW type) provided for power transformer and functioning. b. The Nitrogen Injection Fire Prevention System (NIFPS) functioning. c. Sand pits. d. Fire fighting cylinders expiry. e. Checking of Switchyard equipments, Transformer cooling system, bay and transformer marshalling boxes and cubicles. 7) Disturbance recorder and Fault locator, standalone or built in Main relay, Whether DR is having automatic fault record download facility to a central PC 8) Whether DR is Time Synchronized with the GPS based time synchronizing equipment. 9) Whether substation (765, 400, 220kV) is having Event logger facility (standalone or built-in-SAS) <u>Checklist for Busbar Protection</u> 1) Duplicated Bus bar protection provided for 765kV and 400kV buses or not. 2) In an existing substation if CTs are of different ratios, is biased type bus protection provided. 3) In stations where single bus bar protection is provided, is backup provided by reverse looking elements of distance relays or by second zone	
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			elements of remote end distance relays? 4) In case of GIS where burn through time of SF6 is shorter than remote back up protection is the bus bar protection duplicated irrespective of voltage level? <u>Checklist for Circuit Breakers :-</u> 1) Is breaker fail protection (LBB / BFR) provided for all the Circuit Breakers at 220kV, 400kV & 765kV ratings. 2) For Circuit Breaker connected to line feeder/ transformer feeder, whether operation of LBB / BFR sends direct trip signal to trip remote end breaker? 3) For lines employing single phase auto reclosing, Is start signal from protection trip to LBB / BFR relay is given on single phase basis? 4) Is separate relay provided for each breaker and the relay has to be connected from the secondary circuit of the CTs associated with that particular breaker? 5) Is LBB relay provided with separate DC circuit independent from Group-A and Group-B Protections? 6) Is the LBB initiation provided with initiating contact independent of CB trip relay contact? 7) Is Separation maintained between protective relay and CB trip coil DC circuit so that short circuit or blown fuse in the CB circuit will not prevent the protective relay from energizing the LBB scheme? 8) Are the Current level detectors set as sensitive as the main protection? (Generally setting of 0.2 A is commonly practice for lines and transformers) 9) Is the breaker failure	
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				protection provided with two steps (First stage re trip own CB, Second stage- Trip all associated CBs). This mitigates unwanted operation of breaker failure protection during maintenance and fault tracing. <u>Checklist for communication systems:-</u> 1. In case of PLCC type pf coupling, redundancy of channels, number of channels. 2. In case of OPGW Whether dedicated fibre is being used for Main-I/ Main-II relay or multiplexed channel are being used. <u>Checklist for DC Supply System</u> 1) Two separate independent DC system (220V or 110V)(Source-A and Source-B). 2) Two independent DC system (48V) for PLCC (source-A and source-B). 3) There is no mixing of supplies from DC source-A and DC source-B. 4) Whether the protection relays and trip circuits are segregated into two independent system fed through fuses from two different DC source. 5) Whether Bay wise distribution of DC supply done in the following way: a. Protection b. CB functions c. Isolator/ Earth switch Functions d. Annunciation/ Indications e. Monitoring Functions <u>Additional checks:-</u> 1) Checking of Testing and Maintenance Records. 2) Checking of Earthing system. 3) Testing of UFR and SPS functionality (If available). 4) Checking of other systems like DG set, HVAC system, switchyard and control room	
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				appearance and lighting etc.	
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Procedure for Carrying out above activities:-

A. Routine Testing of Bay:-

- I. Yearly testing schedule is prepared by respective T&C Divisions and the same is communicated to transmission divisions for aligning their important maintenance schedule with testing schedule to avoid multiplicity of shut downs on the same bay.
- II. Shut down is applied by respective transmission division well in advance for OCC approval/ other approvals.
- III. After availing the shut down on proposed dates, testing and other maintenance activities carried out by respective team.
- IV. Review of critical test results should be done at circle/ zone level
- V. If tan delta results show an increase from previous value, frequency of testing should be increased to monitor the trend. Also, variable frequency tan delta may be carried out to interpret the results better. In case of alarming values the bushing in question should be replaced.

B. Extensive Testing of Bay:- Same as routine testing.

C. Protection Audit of Substations:-

Protection audit teams are to be constituted and deputed by director operation/Chief Engineer for carrying out protection audit of different substations. No shut down is required for this activity hence it can be planned at any time.

Corrective action must be planned to rectify protection/maintenance related shortcomings observed during protection audit in a time bound manner.

D and E Checking of Carrier Communication and Protection Coupler:-

These are routine activities carried out by T&C officers and staff posted at different 400 kV and 220 kV sub stations.

F Commissioning of Substation/Augmentation of Bay:- According to the location of upcoming substation, the responsibility of commissioning is assigned to T&C engineer and staff of nearby existing substation, This assignment of responsibility is done by SE/EE after approval from Chief Engineer.

G,H and I MRI, Metering and Testing of Equipment to be Replaced:- Done by assistant engineer in charge in routine way as per requirement. However, joint sealing of large open access consumers is done jointly by EE (T&C), EE(Trans), EE(Test) and EE(Distr) and consumer representative depending upon the voltage level and connectivity agreement.

J Tripping Analysis:- Important tripping events are discussed in various meetings organized time to time by SLDC. Also, tripping and protection related issues are discussed at meetings held at circle level.

K Checking of CVT/PT Voltages:- Done in routine way on monthly basis or on report of any abnormal behavior, If any abnormality is found in all cores, the PT/CVT in question is replaced.

Protection Setting SOP for UPPTCL

Protection System plays an important role in Protection of Transmission Assets as well as in secure Operation of Grid. Protection System also improves power quality. Main Functions of Protection System are:-

- To timely clear Power System faults thereby preventing the extent of damage due to fault current and reduce the chance of collateral damage.
- To prevent the Transmission Equipment from operating under adverse conditions like over voltage, over load and over flux etc.
- To maintain quality of power by load shedding on under frequency.

Proper and coordinated Settings are important for any Protection System. Protection System must be designed and operated in such a way that it is fast enough and has good sensitivity i.e. only faulty section should get isolated from the grid and all other elements must remain in service.

At present almost all of the Transmission Lines of 220 KV and above have Numerical Protections installed on them either by way of Retrofitting of Numerical Protection Relays in place of old Static/Electromechanical Protection Relays or by Replacement of old Control & Protection Panels. Similarly, most of the Transformers and Reactors are also equipped with Numerical Protection Relays.

Protection Settings of 765 KV, 400 KV and 220KV Transmission Lines, Transformers, Reactors etc. are done according to recommendation of Rama Krishna Committee and Guidelines issued by NRPC for Northern Region. In some specific cases like in case of very short over head lines/cables and in Substations having Series Compensation on one or more Lines, these Settings can be modified according to local requirement subject to approval of NRPC. Current Recommendations for Transmission Lines are may be as per following-

A. Distance Protection Setting for 132 KV Lines

1	Z1(I)	80% of Principal Transmission Line (if in Grid), 110% PL in case of Radial Transmission Lines.
	Z1(T)	Instantaneous.
2	Z2(I)	120% of Principal line.
		150% of Principal line (In case of double circuit line).
	Z2(T)	300ms. 400ms, where a long line is followed by a short line.
3	Z3(I)	120% of (Principal Transmission Line+ adjacent longest Transmission Line at other end.) if Substation is in Grid.
		100% of PL+T/F Impedance running in parallel at other end in case of Radially Fed Substations.
	Z3(T)	800ms.
4	Z4R(I)	02Km, to take care of Bus Faults as no Bus Bar Protection is there at 132KV level in UP system.
		160 ms.

B. Distance Protection Setting for 400/220 KV Lines

1	Z1(I)	80% of Principal line
	Z1(T)	Instantaneous.
2	Z2(I)	120% of Principal line.
		150% of Principal line (In case of double circuit line). To take care of under reaching due to mutual inductive coupling between two parallel circuits.
	Z2(T)	350ms. 500ms, where a long line is followed by a short line.
3	Z3(I)	120% of (Principal line+ adjacent longest line at other end.)
	Z3(T)	1000ms For 400 KV and 800ms For 220KV
4	Z4R(I)	(i) In case Bus Bar Protection is available and functional 50% of shortest line at local Bus
	Z4R(T)	500ms For 400 KV and 450ms For 220KV
	Z4R(I)	(ii) In case Bus Bar Protection is not available 2KM
	Z4R(T)	160ms After the Bus Bar Protection is taken back in service, Z4 settings shall be set as per Guidelines.

Remark-

1. Where 220KV Lines are radial then Zone-1 Impedance Setting shall be 110% of PL.
2. If 400 KV or 220KV Bus Bar Protection is not functional, the time Settings of 400/220KV Bus coupler be set at 100 ms (Definite time) and reverse zone Setting (ZR) of each line emanating from that Bus be set 2 Km and time=160ms (Definite Time). When Bus Bar Protection is taken back in service, zone 4 Setting be set as per guidelines i.e. impedance equal to 50% of shortest line at local Bus Bar with time delay of 450ms.

C. Backup Protection

1. Earth Fault Protection with DMT Setting should be enabled in Distance Protection Relays of 132 KV Transmission Lines (Radial). Pick up Current for this may be set at 0.2A with Definite Time Delay of 1.0 sec.
2. In addition to IDMT Settings should be in backup E/F Protection Relays. No Over-Current Setting is recommended in Grid Transmission Lines.
3. In 132 KV Radial Transmission Lines, Over-Current function may be enabled as per local requirement.
4. Back up Earth Fault Protection Relay Settings for 220/132KV Lines:
Following Setting should be adopted in backup E/F relay
 - (i) 220 KV Lines-PS-0.2, TMS-0.2 when distance Protection is healthy.

-0.1 if distance Protection. is faulty.

Time-1.0 sec. If equipped with definite time (DT)

relay

- (ii) 132 KV Lines-PS-0.2, TMS-0.24 when distance Protection is healthy.

-0.12 if distance Protection. is faulty.

Time-1.0 sec. If equipped with definite time (DT)

relay

Remark: In case of CVT failure, Relay should operate on Non-directional feature with same, Settings, for this corresponding PSL will have to be made.

D. Over Voltage Protection: Applicable for 765KV and 400 KV Lines,

➔ Stage 1: Pickup Voltage 110% Time: 5.0 s

Stage 2: Pickup Voltage 140-150% Time: 100 ms.

For Double Circuit 400KV & 765 KV Lines.

- Stage 1: Pickup Voltage 106% -109% Time: 5.0 s
- Stage 2: Pickup Voltage 140-150% Time: 100 ms.

For double circuit lines grading of stage I over voltage Setting must be done in such a way that one circuit trips earlier than the other thereby increasing the load on other circuit and its chances of surviving the over voltage trip.

- E. LBB Protection:** LBB Protection for Transmission Lines may be either distributed type (having Separate LBB Relay for Each Bay) or may be centralized (incorporated in Bus Bar Protection Relay). Following are the Settings adopted for LBB relay:-

PS : 0.2 A ($I=20\% I_n$)
Time : 0.2 sec (Time delay 200 m sec)

- F. SOTF Protection:**

- For SOTF, Over Current Setting should be set 2.0 -2.5A
- When mode is set to Impedance, the Operation Criterion is based on the start of Overreaching Zone from Impedance Zone Measurement (Normally Zone-2).

- G. Broken Conductor:**

- $I_2/I_1=20\%$ and Alarm time =05 seconds.

Only Alarm should be activated for Broken Conductor i.e. no Tripping should occur on Broken Conductor. However, if any Substation is fed through only one EHV line (during initial stage of commissioning) then Broken Conductor Tripping should be provided on that line till other Transmission Lines are connected to that Bus forming a Grid.

- H. Stub Protection :**

- Applicable for Transmission Lines at Substations having one and half Breaker Scheme.
- Recommended Setting is 250% of I_{Base} and Time can be set to 50ms.

Pickup current: 1 A

Time : 0 ms (Instantaneous)

Stub Protection is required only in the substations where dedicated line CT is absent i.e. line current is derived by summation of Bus side CT and Tie CB side CT. This function should be kept disabled in those substations where dedicated line CT is used for distance Protection (in this case line stub section is protected by TEE Protection).

I. Auto Reclose for 400/220 KV Lines:

Single Pole Single Shot
Dead time 1000ms.

Reclaim Time 25.00sec.

CB pole discrepancy Relay Time= 1.5sec. (2.5sec for 1½ scheme)
It should be work in Zone-1 only.

J. Power Swing Block:

→ Block all Zone except Zone-1. (This application applies a blocking signal to the higher impedance zones of distance Relays and allow Zone-1 to trip if the swing enter its operating characteristics. Breaker application is also a consideration when tripping during a power swing. A subset of this application is to block the Zone-2 and higher impedance Zones for a preset time and allow a trip if the detection do not reset.)

K. Load Encroachment: Should be Enabled only in case of 400KV/765KV Transmission Lines

L. Carrier Inter Tripping-

- Applicable for 400/220 KV Transmission Lines.
- It should be permissible under reach at both ends.
- Appropriate PLCC Scheme PUTT/POTT may be adopted as per local requirement in case of long Transmission Lines with series compensation at other end/ remote adjacent substation end.

M. Direct Trip-

Direct Trip Command shall be initiated in following conditions & shall be routed through PLCC/DTPC-

1. Operation of Bus Bar Protection of the Substation.
2. Tripping of Transmission Lines on O/V.

3. Manual Tripping of Transmission Lines while it is in service.

Relay Settings for Transformer Protection:-

These Settings are done as per Manufacturer's Recommendation. Settings may slightly differ from one manufacturer to another. Setting values also differ from one relay to another (in some relays 1st slope starts from origin while in other relays it may start from another value of restraining current I_{r1} . Accordingly, slope will also vary). However, General Setting Guidelines may be as per following:-

- | | |
|---|--------------------------------|
| 1. Differential Pickup current (I_d)- | 0.15 to 0.20 I_n |
| 2. 1 st slope- | 20% to 30% |
| 3. 2 nd slope- | 60% to 80% (generally starting |
| at I | restraining = $4I_n$ |
| 4. 2 nd harmonic blocking- | 15% to 20% |
| 5. 5 th harmonic blocking- | 45% |
| 6. Over flux Setting- | Set as per over flux capacity |
| curve provided by the transformer manufacturer. | |

• REF Relay Setting: -

1. Pickup value = $0.2 \cdot I_n$ Instantaneous (in case of low impedance REF)
2. In case of high impedance REF Protection, the pickup voltage of REF should be calculated by following formula

$$V = I_{sc}(R_{ct} + 2 \cdot R_{lead}) / CTR$$

Where

I_{sc} is maximum Secondary current in A

R_{ct} is CT Secondary resistance in OHM

R_{lead} is lead resistance in OHM

CTR is the ratio of bushing CT used for REF

Protection

• Back Up Relay Settings for transformers:-

Typical IDMT Settings are shown in the following table. These Settings are based on the assumption that CT used for HV and LV side have primary current rating nearest to the transformer full load current (i.e. for 315 MVA

T/F the CT used at HV and IV/LV side have ratio 500/1 and 800/1 respectively).

Sl. No.	Transformers		Dir O/C		Dir Earth Fault	
			PS	TMS	PS	TMS
1	500/315/240, MVA, 400/220/33KV T/F	HV SIDE	1.0	0.30	0.20	0.30
		IV SIDE	1.0	0.25	0.20	0.25
2	200/160/100, MVA, 220/132/11KV T/F	HV SIDE	1.0	0.25	0.20	0.25
		IV SIDE	1.0	0.20	0.20	0.20
3	63/40/20 MVA, 132/33KV T/F	HV SIDE	1.0	0.20	0.20	0.20
		LV SIDE	1.0	0.15	0.20	0.15

High set pickup current Setting for HV and IV side should be kept approximately 80-85% of short circuit current (for Substations having weak In feed, this Setting may further be reduced). Further, High Set pickup value for O/C and E/F are same for the transformer. A definite time delay of 50 to 60ms. must be provided in High set element to prevent chances of transformer getting tripped with line at very close faults on line.

Remark: The Back-up E/F & O/C Settings shall have directional feature. However, in case of failure of CVT Voltage the directional feature must convert into non-directional.

• **OTI & WTI Settings:**

As per guidelines of Manufacture with a safety margin of 05°C in both cases of Alarm & Tripping.

Relay Settings for Reactor Protection:-

These Settings are done as per Manufacturer's Recommendation. Settings may slightly differ from one manufacturer to another. Setting values also differ from one relay to another (in some relays 1st slope starts from origin while in other relays it may start from another value of restraining current Ir1. Accordingly, slope will also vary). However, General Setting Guidelines may be as per following:-